Six math clues · answer key

H2O: Why Water Is Special

Grade 2 math. Read rows as equal groups and add the same number once for each row.

Math 1 · 12

4 + 4 + 4 = 12, so there are 12 ice cubes.

Math 2 · 12

3 + 3 + 3 + 3 = 12, so the array has 12 drops.

Math 3 · 10

2 + 2 + 2 + 2 + 2 = 10, so there are 10 drawn drops.

Math 4 · 15

5 + 5 + 5 = 15, so the array has 15 paperclips.

Math 5 · 10

5 + 5 = 10, so there are 10 ice cubes.

Math 6 · 20

4 + 4 + 4 + 4 + 4 = 20, so the array has 20 drops.

Grade 2 math. Six tasks use rows and columns from 2 through 5, with repeated addition and totals no greater than 25.

Reading & writing · answer key

H2O: Why Water Is Special

Reading · 1

H2O means each water molecule contains two hydrogen atoms and one oxygen atom.

Reading · 2

Rounded drops and the supported paperclip both show the effect of water molecules attracting one another at the surface.

Reading · 3

Because ice floats, a pond can freeze at the top while liquid water remains below for fish.

Writing

Accept a spoken, dictated, drawn, or written explanation that connects one water property to its cause.

H2O names one water molecule: two hydrogen atoms joined to one oxygen atom. Attraction between water molecules helps explain rounded drops and surface tension. Ice has a more open structure, making it less dense than liquid water, so it floats; this helps ponds freeze from the top. Water dissolves many substances, not everything. Stay beside the child for every activity, use only cool water, keep small items away from mouths, and wipe up spills promptly.

Offer dictation: let the child say a complete thought while you write it, then invite them to add a picture, label, or sentence.

Four water clues

H2O: Why Water Is Special

Quiz 1. What is inside one H2O molecule?

One H2O molecule has two hydrogen atoms and one oxygen atom.

Quiz 2. Why can water form rounded beads on wax paper?

Water molecules attract one another, and surface tension helps pull a drop into a rounded bead.

Quiz 3. Why does ordinary ice float in liquid water?

Ice is lighter for its size than liquid water because its molecules form a roomier pattern.

Quiz 4. What happens when sugar dissolves in water?

Water molecules surround the sugar's tiny pieces and spread them through the liquid.

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